

NOMA-based Visible Light Covert Communication with Optical Intelligent Reflecting Surface

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Abstract—Covert communications are essential for modern communication systems, and visible light communication (VLC) offers a promising avenue for covert operations. In this paper, the covert communication for non-orthogonal multiple access (NOMA)-based VLC systems with the aid of optical intelligent reflecting surfaces (OIRS) is investigated. By adopting the random transmit power strategy, the optimization problem is formulated to maximize the effective covert spectral efficiency of the VLC covert (VLCC) system with the detection error probability and connect outage probability constraints. In particular, a closed-form solution of the power allocation ratio is derived with respect to the OIRS configuration, from which the relaxed minorization-maximization algorithm is utilized to solve the OIRS configuration problem. Simulation results validate the effectiveness of the proposed algorithm, and highlight the potential of OIRS to improve the covertness performance in future optical communication systems.

Index Terms—Optical intelligent reflecting surface (OIRS), visible light covert communication (VLCC), non-orthogonal multiple access (NOMA), effective covert spectral efficiency.

I. INTRODUCTION

COVERT communications are becoming increasingly vital in modern communication systems, particularly in environments where privacy is paramount [1]. In traditional wireless communication systems, maintaining the confidentiality of sensitive information is often challenging due to the potential for eavesdropping [1]. Recently, visible light communication (VLC) has emerged as a promising technology for covert communications, primarily due to its inherent characteristics, such as high data rate and immunity to traditional radio-frequency interference [2]. Specifically, VLC utilizes visible light spectrum for communications, making it less susceptible to conventional wireless eavesdropping techniques and providing a unique platform for covert operations [3].

In recent years, power-domain non-orthogonal multiple access (NOMA) has gained significant attention due to its ability to improve the spectral efficiency and support multiple users simultaneously [4]. By allowing users to share the same frequency resource, NOMA achieves better throughput and fairness compared to orthogonal multiple access (OMA) [4]. In particular, superposition coding is utilized at transmitters

and successive interference cancellation (SIC) is adopted at receivers [5], making NOMA an attractive choice for the visible light covert communication (VLCC) [6]. Generally, the core idea leverages dynamic power allocation with randomized adjustments to embed covert user information into power levels of public users, thus ensuring the covert communication.

To further enhance the performance of NOMA-based VLCC systems, optical intelligent reflecting surfaces (OIRS) has shown great potential [7], [8]. OIRS acts as a reconfigurable surface that can modify the optical wireless channel by manipulating each unit individually, enhancing the signal quality or redirecting the transmission path to improve security [9]. OIRS enables fine-grained control over the communication environment [10], providing a viable tool to optimize the secrecy and reliability of VLCC systems [11]. In particular, OIRS can directly increase the covert communication rate by enhancing the channel gain of the legitimate user, or it can interfere with the eavesdropper by altering its channel quality.

In this paper, OIRS is employed in the NOMA-based VLCC system, utilizing the random power scheme to improve the covertness performance. The primary contributions of this study are summarized as follows. Firstly, a comprehensive system model for the NOMA-based VLCC system with OIRS is meticulously established in Section II. Subsequently, in Section III, a detailed formulation of the problem is taken to maximize the effective covert spectral efficiency (ECSE) with the detection error probability (DEP) and connect outage probability (COP) constraints. Thirdly, a closed-form solution for the power allocation ratio is formulated in relation to the OIRS configuration in Section IV, and thereby the OIRS configuration problem is solved by the relaxed minorization-maximization (MM) algorithm. Moreover, the simulation results demonstrate the convergence of the proposed algorithm and the improvements in ECSE compared to other baselines in Section V. Finally, the conclusion is drawn in Section VI.

Notations: a , $\mathbf{a}$, and $\mathbf{A}$ represent the scalar value, vector, and matrix, respectively. $\mathbb{R}_+$ and $\mathbb{B}$ denote the positive set of real numbers and the set of binary numbers, respectively, and $\mathcal{A}$ is the defined set. Moreover, $|\cdot|$, $\sqrt{\cdot}$, ∂ , and $(\cdot)^T$ represent the symbols of absolute value, square root operator, gradient operator, and transpose, respectively.

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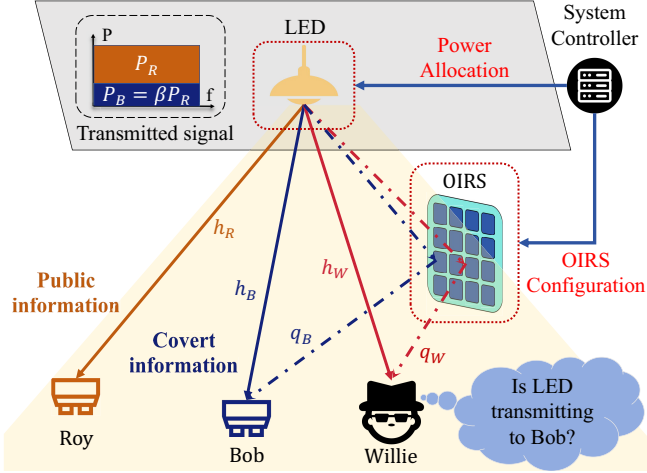


Fig. 1. The system model of the NOMA-based VLCC system with OIRS.

II. SYSTEM MODEL

As depicted in Fig. 1, we consider a NOMA-based VLCC system enhanced by an OIRS. In this setup, a light-emitting diode (LED) transmits information to two legitimate users, Roy and Bob, utilizing the power-domain NOMA. Specifically, the LED openly communicates with Roy who has weak channel quality, while simultaneously conducting covert communications with Bob who has strong channel quality. The communication between the LED and Bob occurs in the presence of a warden, Willie, whose goal is to detect any covert activity. Moreover, an OIRS composed of N units is mounted on the wall, and the indices of these OIRS units are represented by the set $\mathcal{N}$. Without loss of generality, it is assumed that the system controller knows the positions of LED, users, and OIRS units through channel estimation [12].

In the OIRS-aided VLCC system, the transmitted signal arrives at the photodetector (PD) via both line-of-sight (LoS) and OIRS-reflected paths. For LoS paths, the channel gain between the LED and receiver $i \in \{R, B, W\}$ can be modeled by the Lambertian model and denoted as h_i , where $i \in \{R, B, W\}$ denotes one of Roy, Bob, and Willie. For the OIRS-reflected channel, it can be modeled as a signal emitted from a virtual LED [9]. As a result, the OIRS-reflected gain between the LED and receiver $i \in \{R, B, W\}$ through the n -th OIRS unit is represented by $f_{n,i}$ [11], which can be further expressed as a vector $\mathbf{f}_i = [f_{1,i}, \dots, f_{N,i}]^T \in \mathbb{R}_+^{N \times 1}$.

Since the size of OIRS unit is small and the OIRS-reflected signal obeys the Snell's law of reflection [13], the coverage area of the signal reflected by one OIRS unit is limited and can cover only one user in indoor VLC scenarios. Therefore, each OIRS unit forms a reflected channel between the LED and a single receiver. Then, a binary vector $\mathbf{s}_i = [s_{1,i}, \dots, s_{N,i}]^T \in \mathbb{B}^{N \times 1}$, $s_{n,i} \in \{0, 1\}$ is utilized to represent the assignment of OIRS units for receiver $i \in \{R, B, W\}$, where $s_{n,i} = 1$ indicates that the n -th OIRS unit is allocated to the receiver

i . Moreover, the matrix $\mathbf{S} \in \mathbb{R}_+^{N \times 3}$ represents the overall configuration of all OIRS units.

To sum up, the total channel gain for receiver $i \in \{R, B, W\}$ can be obtained by adding the LoS and OIRS-reflected channel gains together, which can be expressed as

$$g_i = h_i + \mathbf{s}_i^T \mathbf{f}_i, \quad i \in \{R, B, W\}. \quad (1)$$

To enable covert communications, a random transmit power scheme is adopted, where the transmit power P_R of the LED to Roy is randomly distributed within the range $[0, P_{\max}]$, with $P_{\max}$ representing the maximum transmit power to Roy. Therefore, the probability density function can be given by

$$f_{P_R}(x) = \begin{cases} \frac{1}{P_{\max}}, & 0 \leq x \leq P_{\max}, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

It is important to note that if the transmit power to Roy were fixed, Willie can easily detect covert transmission by observing any increase in the received power. By introducing variability into the transmit power, however, Willie becomes uncertain whether a rise in the received power is due to a transmission between the LED and Bob or simply a fluctuation in the power of LED directed towards Roy.

When the LED intends to transmit a covert information to Bob, it sends a superimposed signal of the form $\sqrt{P_R} s_R[n] + \sqrt{P_B} s_B[n]$ by power-domain NOMA, where $\sqrt{P_R}$ and $\sqrt{P_B}$ are the power assigned to Roy and Bob by the LED, respectively. Moreover, $s_R[n]$ and $s_B[n]$ represent the n -th normalized information symbols for Roy and Bob, respectively. To ensure the fairness among users, the transmit power allocated to Roy and Bob needs to satisfy that

$$\beta = \frac{P_B}{P_R} \in [0, 1). \quad (3)$$

Therefore, the derived electrical signal at receiver $i \in \{R, B, W\}$ can be written as

$$y_i[n] = \begin{cases} \varsigma \mu g_i P_R s_R[n] + z_i[n], & \mathcal{H}_0, \\ \varsigma \mu g_i P_R s_R[n] + \varsigma \mu g_i P_B s_B[n] + z_i[n], & \mathcal{H}_1, \end{cases} \quad (4)$$

where the null hypothesis $\mathcal{H}_0$ states that the LED did not transmit the covert information, while the alternative hypothesis $\mathcal{H}_1$ means that the LED did transmit the covert information. ς and μ are the electrical-to-optical and optical-to-electrical conversion coefficients, respectively. Moreover, $z_i[n] \sim \mathcal{N}(0, \sigma_i^2)$ is the Gaussian noise at the receiver i .

III. PROBLEM FORMULATION

To formulate the problem, the detection performance of Willie needs to be discussed. To begin with, the average signal power that Willie perceives to be received through the LoS path h_W is given by

$$\tilde{T}_W = \begin{cases} \varsigma^2 \mu^2 h_W^2 P_R + \sigma_W^2, & \mathcal{H}_0, \\ \varsigma^2 \mu^2 h_W^2 (1 + \beta) P_R + \sigma_W^2, & \mathcal{H}_1. \end{cases} \quad (5)$$

The detection performance of Willie can be evaluated by the DEP $\xi = \mathcal{P}_{\text{FA}} + \mathcal{P}_{\text{MD}}$. Specifically, $\mathcal{P}_{\text{FA}}$ and $\mathcal{P}_{\text{MD}}$ represent

the false alarm probability and miss detection probability, respectively. Since Willie detects covert communication based on the channel state information (CSI) of the LoS path, his perceived false alarm probability $\tilde{P}_{FA}$ and miss detection probability $\tilde{P}_{MD}$ are also inaccurate.

First, the perceived false alarm probability that Willie believes in can be written as

$$\begin{aligned} \tilde{P}_{FA} &= \Pr(\zeta^2 \mu^2 h_W^2 P_R + \sigma_W^2 \geq \lambda | \mathcal{H}_0) \\ &= \begin{cases} 1, & \lambda < \sigma_W^2, \\ 1 - \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 h_W^2 P_{\max}}, & \sigma_W^2 \leq \lambda \leq \varpi_{FA}, \\ 0, & \lambda > \varpi_{FA}, \end{cases} \end{aligned} \quad (6)$$

where $\varpi_{FA} = \zeta^2 \mu^2 h_W^2 P_{\max} + \sigma_W^2$. Then, the perceived miss detection probability that Willie believes in can be given by

$$\begin{aligned} \tilde{P}_{MD} &= \Pr(\zeta^2 \mu^2 h_W^2 (1 + \beta) P_R + \sigma_W^2 \leq \lambda | \mathcal{H}_1) \\ &= \begin{cases} 0, & \lambda < \sigma_W^2, \\ \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 h_W^2 (1 + \beta) P_{\max}}, & \sigma_W^2 \leq \lambda \leq \varpi_{MD}, \\ 1, & \lambda > \varpi_{MD}, \end{cases} \end{aligned} \quad (7)$$

where $\varpi_{MD} = \zeta^2 \mu^2 h_W^2 (1 + \beta) P_{\max} + \sigma_W^2$. Therefore, the perceived DEP that Willie believes in can be presented by

$$\tilde{\xi}(\lambda) = \begin{cases} 1, & \lambda < \sigma_W^2, \\ 1 - \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 h_W^2 P_{\max}}, & \sigma_W^2 \leq \lambda \leq \varpi_{FA}, \\ + \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 h_W^2 (1 + \beta) P_{\max}}, & \varpi_{FA} < \lambda \leq \varpi_{MD}, \\ \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 h_W^2 (1 + \beta) P_{\max}}, & \varpi_{FA} < \lambda \leq \varpi_{MD}, \\ 1, & \lambda > \varpi_{MD}. \end{cases} \quad (8)$$

Based on the above expression of $\tilde{\xi}(\lambda)$, Willie would set the optimal decision threshold λ^* as

$$\lambda^* = \zeta^2 \mu^2 h_W^2 P_{\max} + \sigma_W^2. \quad (9)$$

However, the detection of Willie can be confused through the OIRS-reflected channel. In particular, the actual average signal power received by Willie can be given by

$$T_W = \begin{cases} \zeta^2 \mu^2 g_W^2 P_R + \sigma_W^2, & \mathcal{H}_0, \\ \zeta^2 \mu^2 g_W^2 (1 + \beta) P_R + \sigma_W^2, & \mathcal{H}_1. \end{cases} \quad (10)$$

Consequently, the actual DEP at Willie can be expressed as

$$\xi^* = 1 - \frac{\beta h_W^2}{(1 + \beta) g_W^2}. \quad (11)$$

Due to the stochastic nature of the transmit power, the connection outage probabilities (COPs) of the two users also need to be analyzed. According to the SIC process in the power domain NOMA, the generalized signal-to-interference and noise ratio (SINR) that Bob detects Roy's signal $s_R[n]$ can be given by

$$\gamma_{B \rightarrow R} = \frac{\zeta^2 \mu^2 g_B^2 P_R}{\zeta^2 \mu^2 g_B^2 P_B + \tilde{\sigma}_B^2}, \quad (12)$$

where $\tilde{\sigma}_B^2 = e\pi\sigma_B^2/2$ denotes the generalized noise of the covert user Bob [14]. After removing Roy's signal, the generalized SINR when Bob demodulates his own signal can be expressed as

$$\gamma_B = \frac{\zeta^2 \mu^2 g_B^2 P_B}{\tilde{\sigma}_B^2}. \quad (13)$$

Furthermore, when Roy demodulates his own signal, the signal of Bob is treated as interference along with the Gaussian noise, so that the generalized SINR of Roy can be given by

$$\gamma_R = \frac{\zeta^2 \mu^2 g_R^2 P_R}{\zeta^2 \mu^2 g_R^2 P_B + \tilde{\sigma}_R^2}, \quad (14)$$

where $\tilde{\sigma}_R^2 = e\pi\sigma_R^2/2$ denotes the generalized noise of the public user Roy [14]. In addition, the relationship between the spectral efficiency and the generalized SINR can be expressed as $1/2 \log_2(1 + \gamma_i)$, $i \in \{B \rightarrow R, B, R\}$.

Based on the previous discussion, the COP of Bob can be obtained by

$$\begin{aligned} \Xi_B &= 1 - \Pr(R_{B \rightarrow R} \geq R_R^{\text{th}}, R_B \geq R_B^{\text{th}}) \\ &= \begin{cases} 1, & g_B^2 \beta \leq \varpi_{B1}, \\ \frac{\alpha_B \tilde{\sigma}_B^2}{\zeta^2 \mu^2 g_B^2 \beta P_{\max}}, & \varpi_{B1} < g_B^2 \beta \leq \varpi_{B2}, \\ \frac{\alpha_R \tilde{\sigma}_B^2 / \zeta^2 \mu^2 P_{\max}}{g_B^2 - \alpha_R g_B^2 \beta}, & \varpi_{B2} < g_B^2 \beta < \varpi_{B3}, \\ 1, & g_B^2 \beta \geq \varpi_{B3}, \end{cases} \end{aligned} \quad (15)$$

where R_R^{th} and R_B^{th} are the pre-determined SEs from the LED to Roy and Bob, respectively, and $\alpha_R = 2^{2R_R^{\text{th}}} - 1$, $\alpha_B = 2^{2R_B^{\text{th}}} - 1$. Moreover, $\varpi_{B1} = \alpha_B \tilde{\sigma}_B^2 / \zeta^2 \mu^2 g_B^2 P_{\max}$, $\varpi_{B2} = \alpha_B g_B^2 / (\alpha_R + \alpha_R \alpha_B)$, $\varpi_{B3} = g_B^2 / \alpha_R$.

Similarly, the COP of Roy can be expressed as

$$\begin{aligned} \Xi_R &= \Pr(R_R < R_R^{\text{th}}) = \Pr(\gamma_R < \alpha_R) \\ &= \begin{cases} \frac{\alpha_R \tilde{\sigma}_R^2 / \zeta^2 \mu^2 P_{\max}}{g_R^2 - \alpha_R g_R^2 \beta}, & \beta < \frac{1}{\alpha_R}, \\ 1, & \beta \geq \frac{1}{\alpha_R}. \end{cases} \end{aligned} \quad (16)$$

Consequently, the problem of maximizing the ECSE by optimizing the power allocation ratio β and the OIRS configuration matrix $\mathbf{S}$ subject to the given DEP and COP constraints can be formulated as

$$(\mathbf{P0}) : \max_{\mathbf{S}, \beta} R_B^{\text{th}} (1 - \Xi_B) \quad (17)$$

$$\text{s.t. } \xi^* \geq 1 - \kappa, \quad (18)$$

$$\Xi_R \leq \Xi_R^{\text{th}}, \quad (19)$$

$$s_{n,i} \in \{0, 1\}, \quad \forall n \in \mathcal{N}, i \in \{B, R, W\}, \quad (20)$$

$$s_{n,B} + s_{n,R} + s_{n,W} = 1, \quad \forall n \in \mathcal{N}, \quad (21)$$

$$0 \leq \beta < 1. \quad (22)$$

Constraints in (18) and (19) are DEP and COP constraints, respectively, where Ξ_R^{th} is the value of the maximum COP allowed by Roy. Moreover, constraints in (20) and (21) are derived based on the definition of the OIRS configuration

matrix, and constraint in (22) limits the transmit power of Bob to be no greater than that of Roy.

IV. PROPOSED OPTIMIZATION ALGORITHM

In this section, a closed-form solution of the power allocation ratio β with respect to the OIRS configuration is derived first, by which only the matrix $\mathbf{S}$ needs to be considered. Then, the relaxed MM algorithm is utilized to optimize the OIRS configuration.

A. Optimal Power Allocation Ratio

Notice that the expression for DEP ξ^* is given in (11), and so the constraint in (18) can be equivalently transformed into the following form as

$$\beta \leq \frac{\kappa g_W^2}{h_W^2 - \kappa g_W^2}. \quad (23)$$

Moreover, the constraint in (19) can also be equivalently changed according to (16), which can be rewritten as

$$\beta \leq \frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}}. \quad (24)$$

Furthermore, the objective function in (17) is equivalent to minimize the COP of Bob Ξ_B . Observing the expression of the COP for Bob in (15), it can be seen that when $\varpi_{B1} < g_B^2 \beta \leq \varpi_{B2}$, Ξ_B decreases as $g_B^2 \beta$ increases, and when $\varpi_{B2} < g_B^2 \beta < \varpi_{B3}$, Ξ_B increases as $g_B^2 \beta$ increases. Therefore, the optimal value for β should be satisfied $g_B^2 \beta \leq \varpi_{B2}$, which can be expressed as

$$\beta \leq \frac{\alpha_B}{\alpha_R + \alpha_R \alpha_B}. \quad (25)$$

It is worth noting that maximizing the ECSE of Bob is equivalent to maximizing $g_B^2 \beta$. Considering the constraint (22) and the above three inequalities in (23), (24), (25), the optimal value of the power allocation ratio β^* can be obtained as

$$\beta^* = \min \left\{ \frac{\kappa g_W^2}{h_W^2 - \kappa g_W^2}, \frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}}, \frac{\alpha_B}{\alpha_R + \alpha_R \alpha_B}, 1 \right\}, \quad (26)$$

which means that the optimal power allocation ratio needs to be the smallest one of the four values in the above equation. Consequently, the power allocation ratio β^* can be calculated directly by the above equation, then the problem of maximizing the ECSE can be transformed into an optimization only for the OIRS configuration.

B. OIRS Configuration

Since the objective function in (17) is equivalent to maximizing $g_B^2 \beta^*$, the OIRS configuration problem can be obtained based on the derived β^* expression, which is given by

$$(\mathbf{P1}) : \max_{\mathbf{S}} g_B^2 \beta^* \quad (27)$$

s.t. (20), (21).

Algorithm 1 Relaxed MM Algorithm for the ECSE Maximization Problem (P0)

- 1: **Init:** iteration rounds $r \leftarrow 0$, the convergence threshold ϵ , the initial $\Xi_B^{(0)}$ calculated with $\mathbf{S}^{(0)}$ and $\beta^{(0)}$, which are feasible solutions to (P0).
 - 2: **Output:** $\mathbf{S}^*$ and β^* .
 - 3: **repeat**
 - 4: $r \leftarrow r + 1$;
 - 5: Relax the OIRS integer constraint in (20);
 - 6: Solve the relaxed sub-problem (P3-r) by IPM to obtain the relaxed OIRS configuration matrix $\tilde{\mathbf{S}}^{(r)}$;
 - 7: $n \leftarrow 1$;
 - 8: **repeat**
 - 9: $i^* \leftarrow \arg \max_i \tilde{s}_{n,i}^{(r)}$;
 - 10: $s_{n,i^*}^{(r)} \leftarrow 1$;
 - 11: $s_{n,i'}^{(r)} \leftarrow 0, i' \neq i^*$;
 - 12: $n \leftarrow n + 1$;
 - 13: **until** $n > N$;
 - 14: Calculate $\beta^{(r)}$ based on $\mathbf{S}^{(r)}$;
 - 15: Calculate $\Xi_B^{(r)}$ based on $\mathbf{S}^{(r)}$ and $\beta^{(r)}$;
 - 16: **until** $|R_B^{\text{th}}(\Xi_B^{(r)} - \Xi_B^{(r-1)})| \leq \epsilon$;
 - 17: $\mathbf{S}^* \leftarrow \mathbf{S}^{(r)}, \beta^* \leftarrow \beta^{(r)}$.
-

In order to facilitate the solution, it is first necessary to relax the integer constraint in (20) in problem (P1) to a continuous form, i.e.,

$$0 \leq s_{n,i} \leq 1, \quad \forall n \in \mathcal{N}, i \in \{B, R, W\}. \quad (28)$$

Substituting the value of β^* from (26) into the objective function in (27) shows that the objective is a max-min problem. As a result, the original problem (P1) can be reformulated into an equivalent form by introducing a new optimization variable Γ , which can be given by

$$(\mathbf{P2}) : \max_{\mathbf{S}, \Gamma} \Gamma \quad (29)$$

$$\text{s.t. } 2\ln g_B + 2\ln \sqrt{\kappa g_W} - \ln(h_W^2 - \kappa g_W^2) \geq \Gamma, \quad (30)$$

$$2\ln g_B + \ln \left(\frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}} \right) \geq \Gamma, \quad (31)$$

$$2\ln g_B + \ln \frac{\alpha_B}{\alpha_R + \alpha_R \alpha_B} \geq \Gamma, \quad (32)$$

$$2\ln g_B \geq \Gamma, \quad (33)$$

(21), (28),

where the left sides of constraints (30), (31), (32), and (33) are taken logarithmically to simplify the expression, since the value of $g_B^2 \beta^*$ is always larger than zero.

However, constraints (30) and (31) in problem (P2) remain non-convex. Therefore, the MM algorithm is adopted to solve the problem (P2) iteratively. Specifically, in each iteration, the original non-convex constraints are transformed into convex constraints by finding the upper and lower bounds of the non-convex functions, thus facilitating the solution in each

TABLE I
SIMULATION PARAMETERS

Parameter	Value
The number of OIRS units, N	16 or 36
The Lambertian index, m	1
The semi-angle of the FoV, Φ_0	75°
The area of PD, A	1 cm^2
Electrical-to-optical conversion factor, ς	0.5
Optical-to-electrical conversion factor, μ	0.5
The reflection factor, δ	0.95
Pre-determined SE, $R_R^{\text{th}}, R_B^{\text{th}}$	0.5, 1 bps/Hz
DEP constraint, κ	0.25
COP constraint of Roy, Ξ_R^{th}	0.4
Transmit power limit, $P_{\max}$	7 W

iteration. Therefore, the following inequalities are derived by utilizing the Taylor expansion as

$$\ln(h_W^2 - \kappa g_W^2) \leq \ln(h_W^2 - \kappa (g_W^*)^2) - \frac{2\kappa g_W^*}{h_W^2 - \kappa (g_W^*)^2} (g_W - g_W^*), \quad (34)$$

$$g_R^2 \geq \underline{g}_R^2 = (g_R^*)^2 + 2g_R^* (g_R - g_R^*), \quad (35)$$

where g_W^* and g_R^* are fixed values that are calculated from the result $\mathbf{S}^{(r)}$ in the r -th round of iterations. Consequently, the problem (P2) can be relaxed as

$$(\mathbf{P3-r}) : \max_{\mathbf{S}, \Gamma} \Gamma \quad (36)$$

$$\text{s.t. } 2\ln g_B + 2\ln \sqrt{\kappa} g_W \geq \Gamma + \ln(h_W^2 - \kappa (g_W^*)^2) - \frac{2\kappa g_W^* (g_W - g_W^*)}{h_W^2 - \kappa (g_W^*)^2}, \quad (37)$$

$$2\ln g_B + \ln\left(\frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}}\right) \geq \Gamma, \quad (38)$$

$$(21), (28), (32), (33).$$

In this case, the objective function is linear, and all constraints are convex with respect to $\mathbf{S}$ and Γ . Therefore, the problem (P3-r) can be solved iteratively by convex optimization techniques, such as the interior point method (IPM) [15]. During each iteration, the integer constraint in (21) is restored by the minimum distance criterion. This iterative process continues until the difference of ECSE between two consecutive iterations is smaller than a threshold value ϵ , at which point the final OIRS configuration matrix $\mathbf{S}^*$ is obtained, and the power allocation ratio β^* is calculated according to (26). To sum up, the complete algorithmic process is described in **Algorithm 1**.

V. NUMERICAL RESULTS

The numerical results are presented to evaluate the performance of the proposed algorithm. The dimensions of the room are set to $5 \text{ m} \times 5 \text{ m} \times 3 \text{ m}$. Bob, Roy, and Willie are

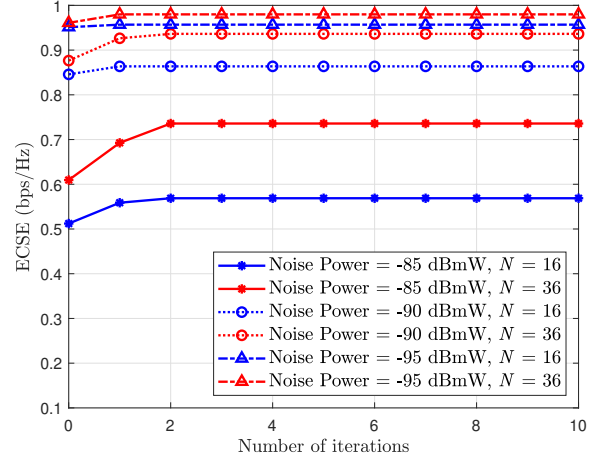


Fig. 2. Convergence of the proposed algorithm under different noise powers and number of OIRS units.

randomly distributed on a plane at a height of 1 m. The LED is set at the center of the roof. Moreover, an OIRS consisting of 16 or 36 units is arranged in a 4×4 or 6×6 configuration on a wall, with (0 m, 2.5 m, 2 m) as the center. Other important parameters are listed in Table I.

The convergence performance of the proposed algorithm across iterations is illustrated in Fig. 2. The results indicate that the algorithm stabilizes within approximately 3 rounds of iteration. Furthermore, a comparison of Bob's ECSE after the convergence reveals a substantial improvement over its initial state, particularly when the noise power at receiver is high or the number of OIRS units is large. For instance, when the noise power is -85 dBmW and $N = 36$, the ECSE improves from an initial 0.60 bps/Hz to nearly 0.73 bps/Hz, representing an increase of approximately 22%.

To demonstrate the benefits brought by the proposed algorithm, it is compared with the following schemes: (1) **Distance greedy** [16]: Each OIRS unit is aligned with the closest user. (2) **Random allocation** [9]: The OIRS configuration matrix $\mathbf{S}$ is determined by the random assignment of each unit. (3) **Without OIRS**: There is no OIRS deployed in the room, the power allocation ratio β can be calculated with the CSI of LoS channels directly.

As shown in Fig. 3, the proposed algorithm achieves the best performance compared to other benchmark schemes. Moreover, it can be seen that the OIRS optimization significantly improves the ECSE performance, especially when the noise power is high. Furthermore, when the noise power is low or the number of OIRS units is large, the performance of all methods approaches 1 bps/Hz, resulting in a smaller gap. This is because the COP of Bob is already very low and the ECSE is close to the limit R_B^{th} , i.e., 1 bps/Hz. In addition, when there is no OIRS, the ECSE remains low, which proves that the introduction of OIRS plays a very important role in the enhancement of ECSE.

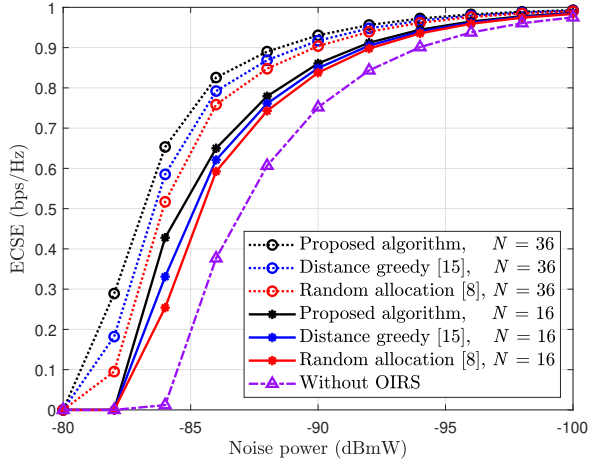


Fig. 3. ECSE versus noise powers compared to different benchmark schemes.

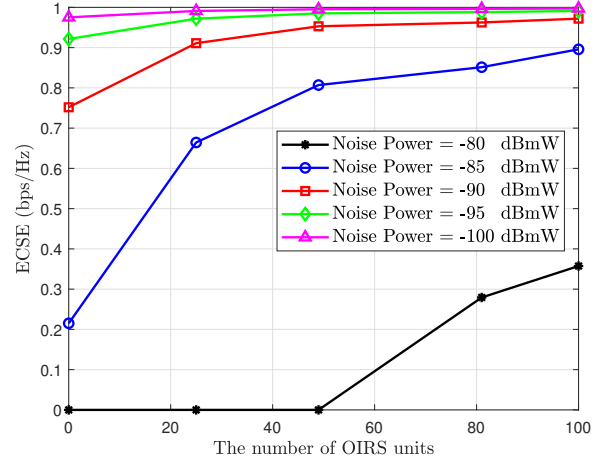


Fig. 4. ECSE versus the number of OIRS units with different noise powers.

Fig. 4 presents the variation in ECSE as the number of OIRS units changes at different levels of noise power. It is evident that increasing the number of OIRS units leads to a significant enhancement in ECSE, with the effect being more pronounced at higher noise power levels. In particular, when the noise power is -85 dBmW, the ECSE is 0.90 and 0.20 for 100 OIRS units and no OIRS, respectively, which is a performance improvement of 4.5 times. These findings align with the results shown in Figs. 2 and 3. In summary, increasing the number of OIRS units serves as an effective approach to improving the ECSE for VLCC systems.

VI. CONCLUSION

In this paper, a NOMA-based VLCC system model with OIRS was established, and the problem was formulated to optimize the ECSE with DEP and COP constraints. To tackle this challenge, a closed-form solution for the power allocation ratio was derived concerning the OIRS configuration. Then, the OIRS configuration was optimized iteratively by a relaxed MM algorithm. Numerical results verified the rapid convergence of the proposed algorithm and its superiority over baselines. Furthermore, simulations demonstrated that increasing the number of OIRS units significantly enhances the ECSE. In conclusion, OIRS can improve the covertness performance of VLCC systems, thereby highlighting OIRS as a promising technology for future optical wireless communications.

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